3) Step-by-step: wire it right for 4K120, VRR, HDR

You'll need: the console, your TV/monitor, and a certified 48 Gbps HDMI 2.1 cable (usually in the box for PS5 Pro/Series X).

1. **Use the right HDMI port** on the TV—look for HDMI 2.1 / "4K120" / "eARC/ARC" labels.
2. **Enable Game Mode/ALLM** on the TV input to reduce latency.
3. **Turn on VRR** on the console and confirm your display supports it. PS5 supports VRR on HDMI 2.1 displays; Xbox supports VRR broadly (including some HDMI 2.0 displays).
4. **Calibrate HDR:**
   - **PS5/PS5 Pro** → Settings ▸ Screen & Video ▸ Video Output ▸ Adjust HDR and follow the on-screen steps (do this per-display).
   - **Xbox Series X|S** → Settings ▸ General ▸ TV & display options ▸ Calibrate HDR for games (uses the system HDR Calibration app).
   - **Windows/PC** → Settings ▸ System ▸ Display ▸ turn on HDR and Auto HDR (if your monitor supports it).

Pro tip: If your game offers both a performance (high-fps) and quality (higher resolution/ray tracing) mode, try performance first on a 120 Hz TV; with VRR, frame pacing smooths out and the game will feel more responsive.

4) Expand storage the supported way

- **PS5 / PS5 Pro:** Add an internal M.2 NVMe SSD that meets Sony's spec; install via the expansion slot and format in Settings when prompted. Sony's support page includes a complete mechanical walkthrough.
- **Xbox Series X|S:** Use official Expansion Cards (e.g., WD_BLACK C50, Seagate) to run Series games from external storage at full speed. Recent buyer's guides confirm both brands are widely sold.
- **Switch 2:** Uses microSD Express cards only (older microSD won't work).

Gaming PCs in 2025

1) The GPU landscape (what's current and why)

- **NVIDIA GeForce RTX 50-series (Blackwell):** Officially launched with RTX 5080 and RTX 5090 early 2025; expect DLSS "4" neural rendering features (multi-frame generation) on 50-series.
- **AMD Radeon RX 9000-series (RDNA 4):** AMD launched the RX 9070 XT/9070 in Feb 2025, emphasizing AI-assisted gaming and next-gen efficiency.
- **CPUs to pair:**
  - **Intel Arrow Lake-S / Core Ultra 200 desktop** platforms (LGA-1851) emphasize lower power and better efficiency than Raptor Lake (2024 launch, still current in 2025).
  - **AMD Ryzen 9000 (Zen 5)** is AM5-compatible and widely available.

2) Upscaling, frame-gen, and latency tech (what to enable)

- **DLSS 3/4 on GeForce:** AI Frame Generation plus NVIDIA Reflex to reduce end-to-end latency. Turn on in supported games; Reflex often lives in the game's video settings.
- **AMD FSR 3 / AFMF on Radeon:** Frame Generation via FSR3 or driver-level AMD Fluid Motion Frames; the latter can be toggled in Adrenalin (and bundles inside HYPR-RX 2).

3) Step-by-step: building or upgrading a gaming PC (GPU-first)

Plan your parts:

- Monitor target (1080p/1440p/4K and refresh rate) → dictates GPU tier.
- Case airflow and PSU wattage (add 30–40% headroom for transient spikes on high-end cards).
- CPU platform (AM5 vs LGA-1851) and PCIe 4.0/5.0 lanes for NVMe.

Install the GPU (brief):

1. Update motherboard BIOS (especially if jumping sockets).